Maria Paul T

ASPIRING DATA SCIENTIST

Contact

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Skills

Generative AI: Prompt Engineering, Hugging Face, Transformers
Deployment, Containerisation Docker, TailScale, SkyPilot, Flask Streamlit,
DBMS: MySQL, PostgreSQL, MongoDB, MariaDB
Analytics & Visualisation: Python, Power BI, MS-Excel
Data Science Libraries: Pytorch, Keras, sklearn, pandas,
Programming: Data Versioning, HTML,CSS,

Certifications

Google IT Support Certification,
Introduction to Linux,
SQL for Data Science,
Machine Learning Foundations: A Case Study Approach ,
Crash Course on Python

Teaching

Teaching Assistant, NPTEL
(Jan 2024- April 2024)
noc24-cs47:

The Joy of Computing Using Python
Prepared weekly assignments & handled the forum for the course having around 60,000 registered learners.

Student Mentor, IITM BS Degree Programme
(Mar 2024- April 2024)
BSMA1004:

Statistics for Data Science 2.
Conducted doubt clearing sessions of 2 hours, 3 times a week for a set of 25 students

Summary

Master's Degree Student building skills in the field of Data Science,

Experience

Project Intern, IIT Jammu | June 2026- Present

Quality-Clustering

- Developed statistical validation framework for quality-bucketing of large text corpora. Implemented topic erasure via linear residualization
- Validated with before/after hypothesis testing (linear probing).
- Used bootstrap Jaccard resampling (20 iterations) to assess cluster stability

Multilingual LLM Evaluation Pipeline

- Built Kubernetes-based evaluation infrastructure (V100 GPU cluster, SkyPilot orchestration) for large-scale multilingual LLM assessment.
- Debugged backend resource allocation and model-serving issues; implemented distributed evaluation across 4,900+ samples.
  Generated reproducible artifact pipeline:
  technical reports + interactive HTML visualizations with embedded plots and sample-level data.

Gaperon mmbert distill repository

- Engineered end-to-end ML pipeline (HuggingFace Hub integration, SkyPilot orchestration) for multilingual model distillation: mmBERT-base → 6-language classifier with independent heads (clarity, coherence, depth, grammar, usefulness, overall). Implemented dataset streaming, checkpoint versioning, and model promotion across branches; optimized for cost and latency across 6 languages (English, Hindi, Tamil, Bengali, Telugu, Malayalam).

Synthia - Synthetic Data Generation Pipeline

- Built production-grade synthetic instruction-tuning data pipeline: raw text → structured triples (context, instruction, response) via self-hosted LLM. Integrated with a private GPU cluster (Tailscale VPN); implemented language-normalized quality filtering (98.9% validation rate) with audit logging.
- Supports pure and code-mixed outputs (Hinglish/Tanglish) with downstream quality tracking—trace any training sample to its generation logic and parameters.

Summer Research Intern, S3 Labs, IIT Bhiliai | May 2025 - July 2025

1. **Technical Image Understanding**
   - Evaluating how Packet Diagrams, Sequence Diagrams, Class Diagrams, State Diagrams, etc are inferenced using LLM's..
2. **Legal Infographic Generation**
   Dataset : IL-TUR(Exploration Lab, IIT Kanpur)
   - Conducted Research Paper Analysis on the Applicability of LLM's in the Legal Domain
   - Generated trial prototypes for generating Infographics like Timelines, Key Arguments
   - **Supervisor: Gagan Raj Gupta, Department of Computer Science and Engineering**

Volunteering

Practical Agentic AI Workshop Series by Prof Raghav Kulkarni, CMI |
August 2026- Present

Conducting Sessions every Sunday from 4PM to 7 PM on Agentic AI.

Coder's High Volunteer :
Volunteered for Workshop Series by Sudarshan Iyenger, IIT Ropar

COVID-19 Response Volunteer:
Caregiver Saathi, May 2021

Hobbies

Room Recreation, Reading.

Others

.GATE DA 2025: Qualified
(March 2025)

Winner Hack Kochi #2:
(November 2019)
IEEE ORE, Kochi Hub

KEAM- Engineering: Rank 333
(August 2018)
Kerala State Entrance Examination

Highest Scorer: Science Stream:
Subject Topper for Maths, Chemistry and Computer Science
(May 2018)

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Projects

1. **Forecasting the Total Marine Fish Production In Kerala - Development & Comparative Analysis of Machine Learning Models**
Machine Learning, Fisheries | Jan 2022- May 2022
- Marine Fish production Data in Kerala (in tonnes) was collected from **CMFRI, Kochi** and data on 9 significant environmental variables from **ICOADS** from 1960 - 2014 .
- Highest predictive power - Random Forest- R2 score 0.87, MAE 0.05, MSE 0.005, RMSE 0.07

2. **VR Industrial Training Simulator | Unity, C# | Jan 2021**
Co-developed a Virtual Reality application simulating boiler feedwater plant control room operations, creating a safe, immersive environment for industrial operator training Collaborated on engineering interactive 3D scenarios and automation workflows in Unity/C#, effectively eliminating the physical risks and costs associated with traditional models.

3. . **Prediction of Crime Categories | Ensembled Learning, Law & Order | August 2024**
- The dataset consisted of 20,000 rows & 20 columns about location, victim demographics, date, time, etc. 6 crime categories were possible for the target variable.
- Data Exploration, Cleaning and Preprocessing created 23 relevant feature columns.
- **LGBMClassifier()** and **XGBoostClassifier()** were used in building a **VotingClassifier()** for which had an accuracy of 0.88080 in contrast to the baseline accuracy of 0.58

4. **Perspective-Aware Sentiment Analysis | Applied Data Analytics | November 2025**

- Built a real-time data pipeline using **Google News RSS** to fetch headlines and established an objective sentiment baseline using the **cardiffnlp/twitter-roberta-base-sentiment** model.
- Evolved the initial single-prompt **GPT-4o-mini persona engine** into a **multi-agent debate framework using LangGraph**, orchestrating s**pecialized agents (Advocate, Context Expert, Skeptic, Judge)** for adversarial reasoning.
- Transformed static polarity labels into explainable, perspective-aware sentiment scores, significantly reducing hallucinations and capturing deep subjective nuances that traditional models miss.

5.. **Opacus - Differential Privacy in PyTorch**
Open Source Contribution, Data Privacy and Data Quality Project | May 2026
- Attempted a Contribution to Opacus, Meta's open-source library enabling differentially private deep learning with minimal code changes. Worked on an issue that had a privacy bug (silent epsilon miscalculations with WeightedRandomSampler)

6. **Structured Multi-Agent Pricing Engine | Agentic AI | November 2026**
- Built a real-time conversational pipeline using the OpenAI Agents SDK and Google Gemini API to automate e-commerce price discovery, orchestrating specialized agents (Triage, Researcher, Calculator) for seamless task routing.
- Evolved the initial single-agent prompt into a multi-agent orchestration framework using Pydantic for strict schema validation and SQLite for persistent session memory, ensuring type-safe data handoffs and context-aware interactions.
- Transformed unstructured user queries into structured, actionable pricing records, significantly reducing model hallucinations and preventing infinite tool loops through rigorous prompt constraints and state-tracking.

Education.

Chennai Mathematical Institute, **Master of Science**
Data Science, Aug 2024 - May 2026